Lukas Klein

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Work Experience

École Polytechnique Fédérale de Lausanne (EPFL) *06/2025 – Today*
Postgraduate Researcher – In the AI for Molecular Medicine Group with Prof. Bunne.
- Collaborator in the National AI Initiative for Precision Oncology Innosuisse Flagship Grant

University Hospital Zürich (USZ) *06/2025 – Today*
Postgraduate Researcher – In the Department of Oncology with Prof. Wicki.

German Cancer Research Center (DKFZ) *08/2021 – 06/2025*
Graduate Researcher - Supervised by Dr. Jäger (Google DeepMind) and Prof. Maier-Hein (DKFZ).

Eidgenössische Technische Hochschule (ETH) Zürich *02/2021 - 08/2021*
Research Intern - Institute for Machine Learning supervised by Prof. Buhmann.

Camelot ITLab *01/2019 - 08/2019*
IT Consultant - Machine Learning and cloud solutions in the pharma and chemical industry.

Camelot ITLab *01/2018 - 01/2019*
Thesis and Working Student - Bachelor Thesis and Data Science consulting.

Microsoft *06/2017 - 09/2017*
Intern - National Technology Office supervised by Thomas Langkabel.

Education

Eidgenössische Technische Hochschule (ETH) Zürich *2021 - 2025*
Ph.D. in Computer Science – Supervised by Prof. Buhmann, Prof. Krause, and Dr. Jäger.
- External group member in the IVIA Lab of Prof. El-Assady.
- Collaboration with the MIT-IBM Watson AI Lab supervised by Dr. Strobelt.

Helmholtz International Graduate School for Cancer Research *2021 - 2025*
DKFZ International Scholarship Award - Acceptance rate of 2.7% in 2022.
- Completed the DKFZ International PhD Program for Cancer Research.

University of Oxford *2022*
OxML Summer School for Machine Learning in Healthcare.

University of Geneva *2019 - 2021*
M.Sc. in Statistics – Final grade: Very Good. Master Thesis and Exchange Semesters at ETH Zürich.

University of Mannheim *2015 - 2018*
B.Sc. in Economics - Final grade: Good. Exchange Semesters at the Norwegian School of Economics.

Professional Services

Invited Talks and Seminars

11/2024 [Talk] **Tufts University** Visual Analytics Lab (VALT)
10/2024 [Talk] **MIT-IBM Watson AI Lab**
10/2024 [Talk] **Harvard University** Harvard Data Science Initiative (HDSI)
10/2024 [Talk] **Harvard University** Visual Computing Group
10/2024 [Talk] **MIT** Visualization Group
06/2024 [Talk] **IBM Research** AI for Scientific Discovery Group
12/2023 [Talk] **NeurIPS** XAI in Action 2023 Workshop
06/2023 [Talk] **TU Munich** Computational Imaging and AI in Medicine Group

06/2023 [Talk] **Helmholtz AI Conference** Interdisciplinary discussion **"**When do we blindly trust in AI?"
06/2023 [Seminar] **Helmholtz Imaging & Helmholtz AI Conference**
04/2023 [Talk] **DESY Research Centre**
09/2022 [Seminar] **Helmholtz Information & Data Science Academy (HIDA)** HIDA Summer School
05/2022 [Seminar] **HIDSS4Health Graduate School** Seminar for Analysing Medical Images Using ML

Conference Reviewer

Neural Information Processing Systems (**NeurIPS**)
International Conference on Machine Learning (**ICML**)
International Conference on Learning Representations (**ICLR**)
Eurographics Conference on Visualization (**EuroVis**)
Including various Workshops.

Teaching

2023-2024 Advanced Topics in Machine Learning (ETH Zürich)
2022-2024 Biomedical Imaging and Radiation Oncology (University of Heidelberg)

Community Service

2022 – 2025 **Advisory Committee of the Oxford ML Summer School**
AI for Global Goals.

2016 – 2017 **Speaker for Public Relations**
AStA (Student Council) at the University of Mannheim.

2015 – 2016 **Member of the Student Parliament and Deputy Member of the Information Committee**
University of Mannheim. Representative for Refugee Projects.

Publications

Google Scholar: https://scholar.google.com/citations?user=sxpuZg4AAAAJ

2025 **Explainable AI-based analysis of human pancreas sections identifies traits of type 2 diabetes**
Lukas Klein, Sebastian Ziegler, Yanni Morgenroth, Eyke Schoeniger, Michele Solimena, Felicia Gerst, Robert Wagner, Fabian Isensee, Klaus Maier-Hein, Paul Jäger. *Venue: Under review at Nature Communications.*

2025 **nnActive: A Framework for Evaluation of Active Learning in 3D Biomedical Segmentation**
Carsten T. Lüth, Jeremias Traub, Kim-Celine Kahl, Till J. Bungert, **Lukas Klein**, Lars Krämer, Paul F Jaeger, Fabian Isensee, Klaus Maier-Hein. *Venue: TMLR.*

2025 **Finally Beating the Random Baseline: A Simple and Effective Active Learning Method for 3D Biomedical Segmentation**
Carsten T. Lüth, Jeremias Traub, Kim-Celine Kahl, Till J. Bungert, **Lukas Klein**, Lars Krämer, Paul F Jaeger, Klaus Maier-Hein, Fabian Isensee. *Venue: Under review at NeurIPS 2025.*

2024 **Why context matters in VQA & Reasoning: Semantic interventions for VLM input modalities**
Lukas Klein, Kenza Amara, Carsten Lüth, Antonio Rodríguez, Hendrik Strobelt, Mennatallah El-Assady, Paul Jäger. *Venue: Under review at WACV.*

2024 **Navigating the Maze of Explainable AI: A Systematic Approach to Evaluating Methods and Metrics**
Lukas Klein, Carsten Lüth, Udo Schlegel, Till Bungert, Mennatallah El-Assady, Paul Jäger. *Venue: NeurIPS 2024.*

2024 **Enhancing predictive imaging biomarker discovery through treatment effect analysis**
Shuhan Xiao, **Lukas Klein**, Paul Jäger, Jens Petersen, Klaus Maier-Hein. *Venue: WCAV 2025.*

2024 **Interactive Semantic Interventions for VLMs: Breaking VLMs with Human Ingenuity**
Lukas Klein, Kenza Amara, Carsten Lüth, Antonio Rodríguez, Hendrik Strobelt, Mennatallah El-Assady, Paul Jäger. *Venue: NeurIPS 2024 Safe Generative AI Workshop* and *Causality & Large Models Workshop.*

2024 **Explainable AI-based analysis of human pancreas sections detects traits of type 2 diabetes**
Lukas Klein, Sebastian Ziegler, Yanni Morgenroth, Eyke Schoeniger, Michele Solimena, Felicia Gerst, Robert Wagner, Fabian Isensee, Klaus Maier-Hein, Paul Jäger. **Venue:** *NeurIPS 2024 IAI Workshop.*

2023 **Discovering Process Dynamic for Scalable Perovskite Solar Cell Manufacturing with Explainable AI**
Lukas Klein, Sebastian Ziegler, Felix Laufer, Charlotte Debus, Markus Götz, Klaus Maier-Hein, Ulrich W. Paetzold, Fabian Isensee, Paul Jäger. *Journal: Advanced Materials (IF: 30).* ***This work has been covered i.a. by The Independent.***

2023 **Understanding Solar Cell Manufacturing with Explainable AI**
Lukas Klein, Sebastian Ziegler, Felix Laufer, Charlotte Debus, Markus Götz, Klaus Maier-Hein, Ulrich W. Paetzold, Fabian Isensee, Paul Jäger, *Venue: NeurIPS 2023 Workshop on XAI in Action* ***(Oral, Top 3 Paper).***

2023 **Navigating the Pitfalls of Active Learning Evaluation: A Systematic Framework for Assessment**
Carsten T. Lüth, Till J. Bungert, **Lukas Klein**, Paul Jäger. *Venue: NeurIPS 2023.*

2023 **Semi-supervised automated Gleason Grading on WSI**
Maximilian Fischer, Lars Krämer, Sebastian Ziegler, **Lukas Klein**, Paul Jäger, Marco Nolden, Peter Neher, Fabian Isensee, Klaus Maier-Hein, *Venue: ECDP 2024 (Oral).*

2023 **A Call to Reflect on Evaluation Practices for Failure Detection in Image Classification**
Paul Jäger, Carsten T. Lüth, **Lukas Klein**, Till J. Bungert. *Venue: ICLR 2023* ***(Oral, Top 1% Paper).***

2022 **From Correlation to Causation: Formalizing Interpretable Machine Learning as a Statistical Process**
Lukas Klein, Mennatallah El-Assady, Paul Jäger. *Venue: IJCAI 2022 Workshop on XAI.*

2022 **Improving Explainability of Disentangled Representations using Multipath-Attribution Mappings**
Lukas Klein, João Carvalho, Mennatallah El-Assady, Paolo Penna, Joachim M. Buhmann, Paul Jäger. *Venue: MIDL 2022.*

Languages & Skills

Languages: German (Native), English (Fluent), French (Basic)

Programming Languages: Python, R, Bash, Git, LaTeX, SQL, Julia

Libraries: PyTorch, Lightning, Hydra, Numpy, SciKit, Transformers, XGBoost, JAX, Lme4, Tidyverse

Software & Platforms: Linux (Ubuntu, CentOS, Debian), GitHub/GitLab, IBM Spectrum LSF, Docker, Kubernetes, RunAI, QuPath, Weights & Biases, Nvidia Jetson IoT Platform, CMake, Azure (ML Studio, OpenAI Services), Google Cloud (Cloud Build & Run, Vertex AI), SAP Analytics Cloud, OpenAI API Studio, NCI Genomic Data Commons & TCGA